jwselect: Quick User Guide

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Select optional preperiod restrictions and estimate treatment effects by minimum distance (MD), using cohort–time estimates and their joint covariance. This guide covers the ordinary estimation workflow.

1 Install

The main command requires Stata 16.1 or later. Extract the current archive and replace the path below with the folder containing `jwselect.pkg`:

```
net install jwselect, from("C:/path/to/jwselect") replace
which jwselect
findfile _jwselect_mata.do
help jwselect
```

After an update, save your work and restart Stata. To use `jwselect` directly after `jwdid`, first install its dependencies:

```
ssc install require, replace
ssc install ftools, replace
ssc install reghdfe, replace
ssc install hdfe, replace
net install jwdid, ///
    from("https://raw.githubusercontent.com/friosavila/stpackages/main") replace
```

Run commands containing `///` together from the Do-file Editor. An optional installation check, in a fresh Stata 17 session, is `do run_release_tests.do` from the extracted package folder. The test clears data and estimates; save your work first.

2 A typical analysis

Replace the variable names below with yours. Code never-treated units as `first_treat=0`; treated units carry their first treatment date.

```
jwdid y x1 x2, ivar(id) tvar(year) gvar(first_treat) ///
    never method(regress) cluster(id)
estimates store JW_SOURCE

* Maintained-baseline estimate, with no optional restrictions
jwselect, select(manual) postwindow(0 2)
estimates store JW_BASE

* Default: cohort-wise Wald-BIC, followed by joint MD
estimates restore JW_SOURCE
jwselect, select(wbic) candidates(nested) postwindow(0 2)
estimates store JW_WBIC
```

Restore `JW_SOURCE` before every new direct `jwselect` run. Each successful run replaces Stata's estimation results. A bare `jwselect` only redisplays the current results. Here, `postwindow(0 2)` averages the first three treatment periods within each cohort.

Supported direct input. Use a balanced, regularly spaced panel, unweighted linear estimation, full cohort–time effects, clustering by `id`, and numeric covariates constant within each unit. The default `vce(unconditional)` includes covariate-averaging uncertainty and requires `method(regress)`. For other designs, use the matrix interface on page 5.

3 Choose the design and preperiod restrictions

Baseline and treatment-effect horizon

The baseline is maintained, never selected or pretrend-tested. Optional dates add restrictions. For annual data, g is the cohort's treatment date and q the earliest input treatment date. With `baseline(common)` and no anticipation, eligible dates extend through $g-1$, excluding the baseline $q-1$.

Option	Meaning
`baseline(cohort)`	Baseline $g-1$; the default.
`baseline(common)`	Baseline $q-1$ for every cohort.
`anticipation(1)`	Allow one anticipation period: use $g-2$ or $q-2$ and exclude potentially affected optional dates. Default: 0.
`postwindow(0 0)`	Immediate effect only, for each cohort.
`postwindow(0 2)`	Average over event times 0, 1, and 2.

Without `postwindow()`, all available post-treatment cells are used. Every requested date must exist for every cohort. For spacing $h =$ `gap()` and anticipation δ, the baseline is $g-h(1+\delta)$ or $q-h(1+\delta)$; event time is $(t-g)/h$. Direct input obtains the spacing from `jwdid`. The two commands use different anticipation conventions; do not copy the `jwdid` anticipation value mechanically into `jwselect`.

Baseline only, manual choice, or automatic selection

Option	What it does
`select(manual)`	Use only the maintained baseline unless `prematrix()` is supplied.
`select(wbic)`	Default. Select within each cohort; add `wbicscope(joint)` for joint selection (next page).
`select(focused)`	Balance estimated bias and variance for a chosen target; see page 4.

For manual selection, list optional dates in a two-column matrix: *cohort, calendar time*. For example, if cohorts 4 and 6 and these dates are present in your data:

```
matrix P = (4,1 \ 4,2 \ 6,3 \ 6,4)
estimates restore JW_SOURCE
jwselect, select(manual) prematrix(P) postwindow(0 2)
```

Do not list the baseline. Dates need not be consecutive. Manual and Wald–BIC use the *union* of the chosen restrictions in joint MD: one cohort's restrictions can affect other estimates through covariance. A cohort absent from `P` contributes no optional restrictions of its own.

Choose the candidate family explicitly

With optional dates $\{1,2,3\}$, `candidates(nested)` searches $\varnothing, \{3\}, \{2,3\}, \{1,2,3\}$. `candidates(all)` searches all eight subsets, including $\{1,3\}$. With K optional dates, these families have $K+1$ and 2^K candidates. The package default is `nested`; request `all` explicitly for exhaustive search. Baseline only is included in both families.

The limits are `maxcand(4096)` per cohort and `maxjoint(4096)` for a joint WBIC or aggregate-focused family. Exceeding a limit causes an error; no candidates are silently dropped. All-subset searches grow quickly; raise a limit only if you intend the larger family.

Aggregation. The default gives cohorts equal weight, divided equally among their included post periods. `aggregate(equalcell)` weights included cells equally.

Joint Wald–BIC selection (new in 0.4.0)

Use one joint criterion to choose restrictions across all cohorts:

```
estimates restore JW_SOURCE
jwselect, select(wbic) wbicscope(joint) ///
    candidates(nested) postwindow(0 2)
estat selection
estat scores
estimates store JW_JOINT
```

The joint criterion is $W_A - r_A \log N$: the joint pretrend Wald statistic, minus the number of restrictions times the log of the independent-unit count. It uses covariance across cohorts and dates, then estimates all effects by joint MD. The default `wbicscope(cohort)` selects within cohorts before joint MD; these two rules can choose different sets.

The joint family combines every local candidate with every candidate in other cohorts. With K_g optional dates in cohort g, there are $\prod_g (K_g + 1)$ nested configurations or $2^{\sum_g K_g}$ all-subset configurations. Even nested joint search may exceed `maxjoint(4096)`. A prespecified `jointmatrix(JM)` can supply a smaller family: one binary joint mask per row, with columns in `e(hmap)` order. Obtain that map with `estat design`; see `help jwselect` for details. Fix the family before inspecting outcomes. Raise a cap only for a computationally feasible family.

Joint pretrend tests

After any successful `jwselect` fit:

```
* All eligible optional preperiod restrictions, jointly
estat pretrend
return list

* Manual/Wald-BIC selected set: diagnostic after selection
estat pretrend, selected
```

For focused results, explicitly choose an output instead:

```
matrix list e(output_map)
estat pretrend, selected output(1)
```

`output()` means a result-row number and is allowed only with `selected`. Manual/Wald–BIC selected tests default to row 1, since all rows share the same joint restriction set. For a prespecified subset, use `estat pretrend, prematrix(P)`; `P` has the same two-column cohort/date format as manual selection. Do not combine it with `selected`.

The test reports Wald, the equivalent MD J, degrees of freedom, and a chi-square reference p-value. J is evaluated through the Wald identity under the compatible covariance-weighted MD model, not by an independent refit. `r(table)` has columns `df Wald J pvalue`; separate scalars include `r(chi2)`, `r(J)`, `r(df)`, and `r(p)`.

Interpretation. A prespecified test uses the chi-square reference under the maintained model. A p-value for an outcome-selected set is a diagnostic, not an automatic correction for selection. Rejection concerns optional restrictions; the maintained baseline is never tested. For a focused aggregate of cohort-selected effects, the test uses the union of their restrictions; J refers to that set's compatible joint MD model. An empty set has zero degrees of freedom and no p-value.

4 Focused estimates and confidence intervals

Request cohort-focused estimates, their aggregate, and numerical inference:

```
estimates restore JW_SOURCE
jwselect, select(focused) focus(cohort) ///
    candidates(all) postwindow(0 2) level(95)
estat inference
```

Option	Selection target and reported estimates
`focus(cohort)`	Select for each cohort's mean effect; also report the aggregate of the resulting cell estimates.
`focus(aggregate)`	Select a joint restriction configuration for the aggregate. Cohort results use that same selected model.
`focus(both)`	Package default. Report cohort-focused effects, their aggregate, and a separately aggregate-focused estimate.

For point estimates only, add `inference(none)`; intervals and p-values will be missing.

How to read focused inference

The default `inference(auto)` repeats selection inside joint Gaussian draws and searches over plausible preperiod violations. Use the reported intervals and p-values. `se_fixed` is a fixed-selection diagnostic; do not build ordinary normal intervals from it. Intervals are marginal for each effect, not simultaneous bands, and need not be narrower than baseline-only intervals.

Search budget: the controls that matter

Option	Default	Meaning
`optmaxeval()`	400	Evaluations per objective, shared across starts.
`optstarts()`	4	Starting points per objective.
`optreps()`	1,999	Common Gaussian draws used during search.
`reps()`	19,999	Independent draws used to check discovered points.
`opttol()`	0.02	Stopping mesh size in standardized coordinates.
`seed()`	12,345	Reproducible simulation seed.

There are four objectives per effect: two interval bounds and two p-value tails. Each start receives 100 evaluations by default. **"Search budget exhausted" means at least one start reached its allowance.** The command returns the best points found; this is not a Stata failure. Inspect `budget_exhausted` and `converged_starts` in `estat inference`.

First increase the evaluation budget while retaining the same design:

```
estimates restore JW_SOURCE
jwselect, select(focused) focus(cohort) ///
    candidates(all) postwindow(0 2) level(95) optmaxeval(1600)
estat inference
```

Compare intervals and p-values with the earlier run. More starts, more draws, and another seed provide further checks. Increasing `reps()` alone does not resolve budget exhaustion. Converged searches do not certify global extrema.

The defaults `gamma(.01)` and `mcerr(.001)` allocate error to the nuisance region and simulation checks. See the full guide before changing them.

5 Read, save, and supply results

Inspect and save a completed run

Each cohort row averages its included post periods. The aggregate uses the chosen cell weights. `e(output_map)` identifies the actual cohort behind each row; an aggregate has cohort value zero.

```
jwselect                    // Redisplay the current results
estat selection             // Candidate sets and selected restrictions
estat design                // Baselines and cohort-time maps
estat cells                 // Individual cohort-time estimates
matrix list e(output_map)
estimates save "jwselect_results.ster", replace
```

These commands follow a successful `jwselect` call. Store or save each result you want to compare before restoring `JW_SOURCE`.

Stored result	Use
`e(table)`	Main table: estimate, `se_fixed`, interval bounds, and zero-effect p-value.
`e(b)`, `e(ci)`, `e(pvalue)`	Estimates, confidence intervals, and p-values separately.
`e(selection)`	Restrictions used by each output; absent when no optional preperiods exist.
`e(V)`	Joint covariance for manual and Wald–BIC results.
`e(V_fixed)`	Fixed-selection covariance; diagnostic for focused estimates.

Prespecify manual restrictions, candidate families, and target weights. Manual intervals treat the chosen restrictions as fixed. Wald–BIC intervals use a pointwise oracle approximation under the method's assumptions; they are not uniformly valid near violations. Focused results do not post an ordinary `e(V)`. Do not use `e(V_fixed)` with conventional `lincom` or `test` as if it adjusted for focused selection.

Alternative input: coefficient and covariance matrices

Use this route for separately validated cohort–time estimates and their full covariance, including designs outside the direct adapter. It requires no `jwdid`.

```
jwselect, bmat(B) vmat(V) cells(C) nunits(500) ///
    gap(1) select(wbic) candidates(all) postwindow(0 2)
```

Input	Required contents
`B`	Row or column vector of unpooled treated-cohort–time estimates.
`V`	Full sampling covariance of `B`, including covariance across cohorts and dates. Supply V, not NV.
`C`	Matching rows: *cohort, calendar time*. Include the normalization cells and only positive treated-cohort codes.
`nunits()`	Number of independent sampled units, used in the Wald–BIC penalty; replace 500 with your count.
`gap()`	Period spacing, with default 1.

All design and selection options above apply. Repeat the matrix arguments for each new specification. The covariance convention is `vce(supplied)`. Pooled event-time output generally lacks the required cohort–time information. Normalization rows may be zero; do not insert unidentified effects as estimated zeros. Validate the inputs and covariance before using this interface.